LAB PYTHON PROGRAMS
CSA1718-ARTIFICIAL INTELLIGENCE

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3. Write a program to solve water jug problem

PROGRAM:

```
def water_jug():
 jug1, jug2 = 0, 0
 capacity1, capacity2 = 4, 3
 goal = 2

 while jug1 != goal and jug2 != goal:
 if jug1 == 0:
 jug1 = capacity1
 elif jug2 < capacity2:
 pour = min(jug1, capacity2 - jug2)
 jug1 -= pour
 jug2 += pour
 else:
 jug2 = 0

 print("Jug 1:", jug1, "Jug 2:", jug2)

 print("Goal reached!")

water_jug()
```

OUTPUT

```
Jug1: 4 Jug2: 0
Jug1: 1 Jug2: 3
Jug1: 1 Jug2: 0
Jug1: 0 Jug2: 1
Jug1: 4 Jug2: 1
Jug1: 2 Jug2: 3
Goal Reached: 2 litres
```